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Seminar Cognitive Systems 2: Behavior Control

Seminar Task

Problem KS2.1 [P-551.S1.EN] – Sliding Block Puzzle

A sliding block puzzle is a (often square) puzzle in which the individual pieces are interlocked and bordered by a frame, so that they cannot be placed, but can only be slid past one another. In an $N \times N$ puzzle there are $N^2 - 1$ blocks and a free field that allows horizontal or vertical movement of adjacent blocks. The goal of the game is to arrange the blocks in ascending order.

Seminar Task: Write a program that solves a 3×3 sliding block puzzle using at least two different search algorithms:

1. an uninformed algorithm, e.g., breadth- or depth-first search, and
2. A* search including a permissible heuristic.

Compare the performances of the search algorithms you implemented and discuss your observations at the end of your talk!

Requirements: The start and target configurations (including the position of the free field!) of the puzzle are

start:

8	7	6
5	4	3
2	1	

target:

	1	2
3	4	5
6	7	8

In order to compare algorithm performances, your program must count and display

1. the number of node expansions,
2. the maximum number of nodes held in memory at any point of time, and
3. the number of moves needed to solve the puzzle.

The choice of programming language is free. However, you must not use any libraries which contain implementations of the search algorithms! Prepare a 10-minute presentation of the theory of the implemented search algorithms and your results!

A maximum of 10 points may be awarded: 4 for the program, and 6 for the presentation and discussion. Points awarded for the seminar will be added to the points earned in the exam.

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